



Evaluation of Autoencoders-based strategies for dimensionality reduction of high-dimensional financial data

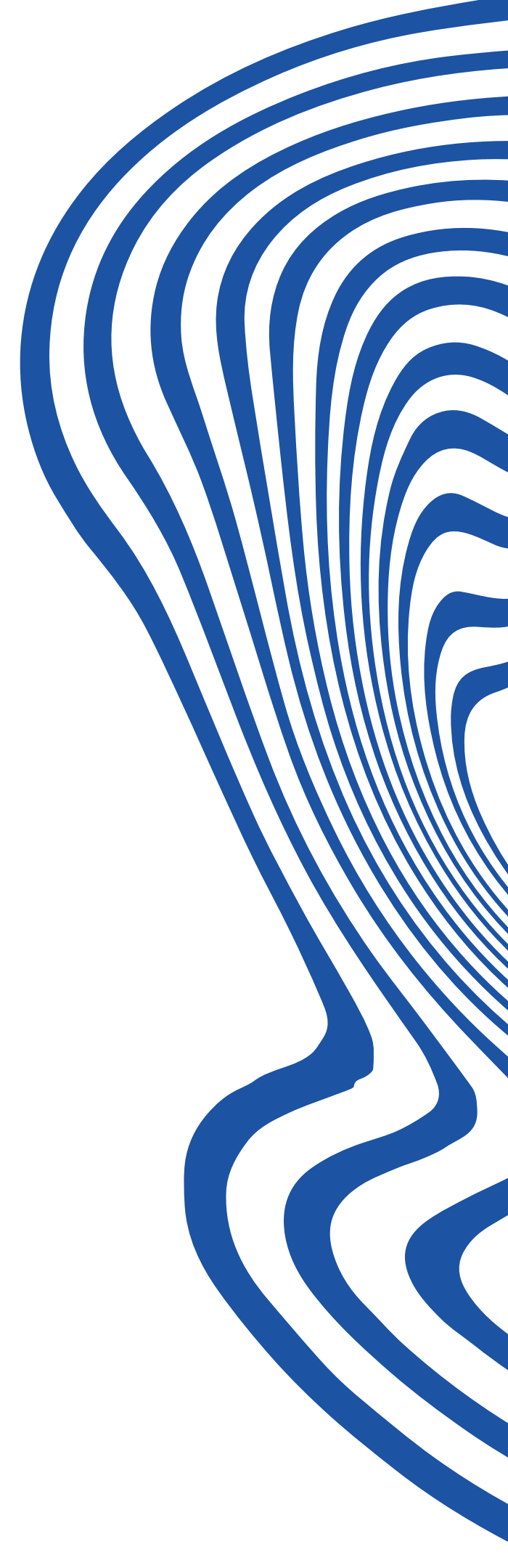
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Research Goals

- Investigate the performance of different AE architectures for dimensionality reduction task
- Compare that performance to PCA
- Analyze additional applications of the latent space representation



Autoencoder architectures in scope

Main models of interest

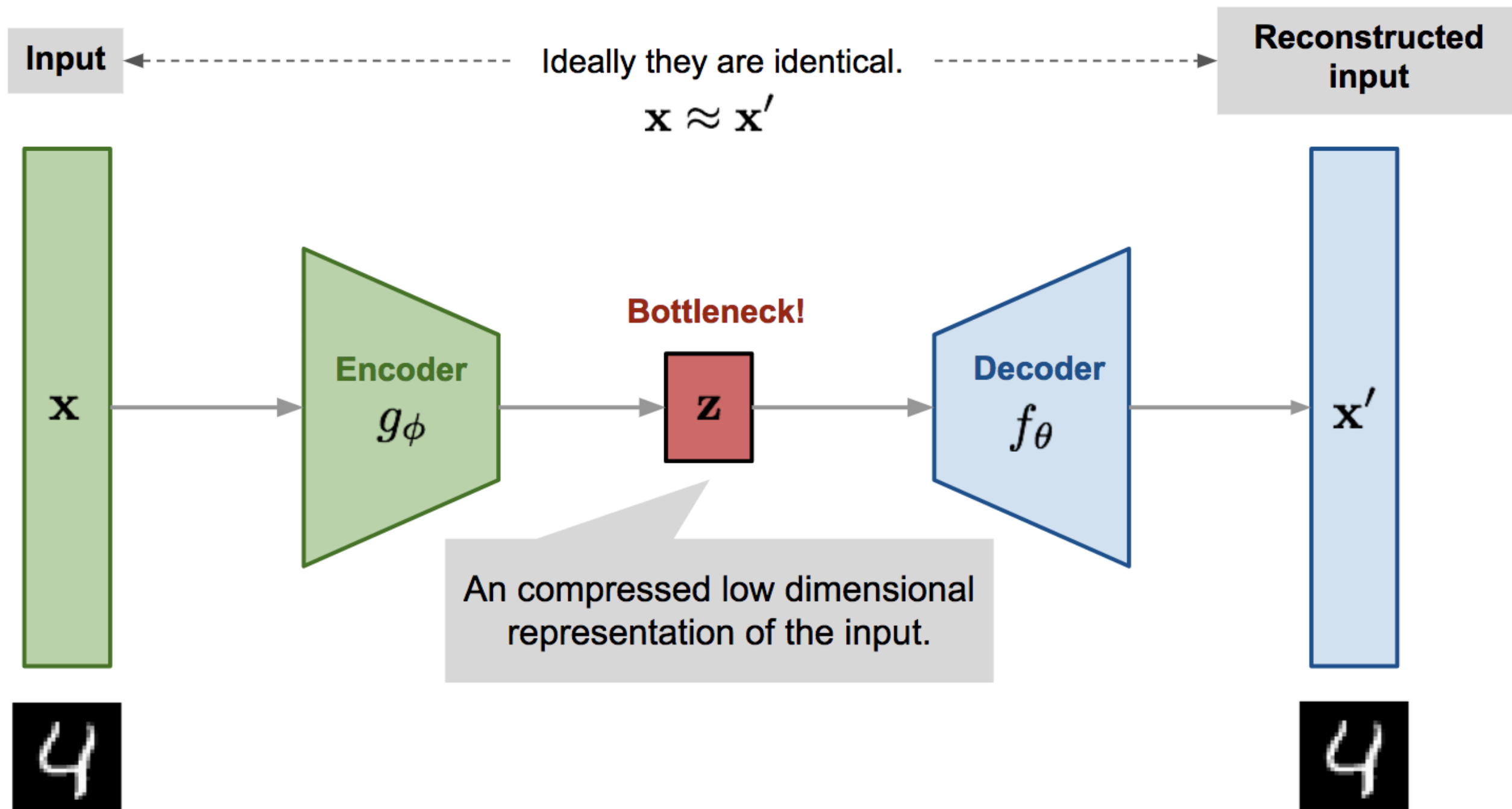
- PCA
- Regular autoencoder
- Variational Autoencoder (VAE)
- β -VAE
- VAE with t-student prior/posterior assumptions

Additional models

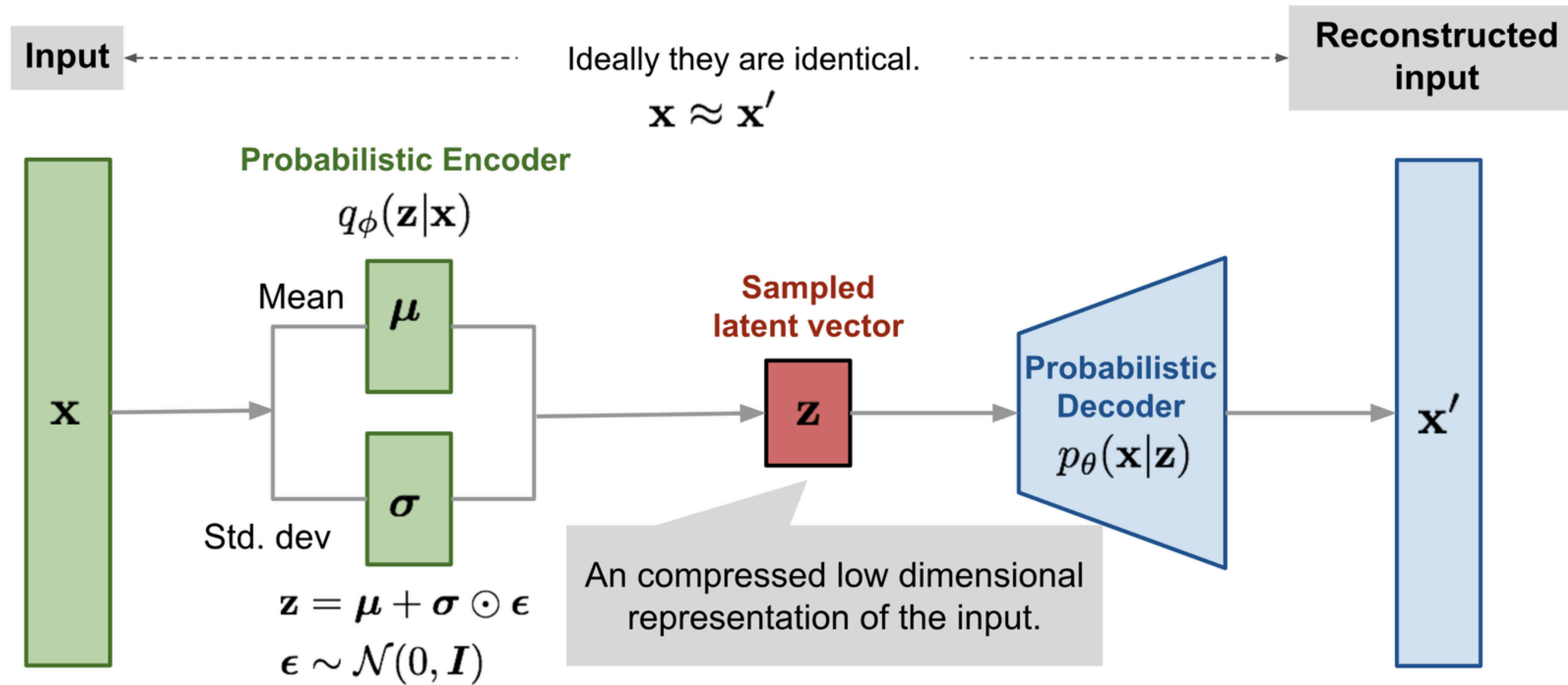
- VAE with gaussian copula-based prior
- VAE with vine copula prior



Autoencoder



VAE

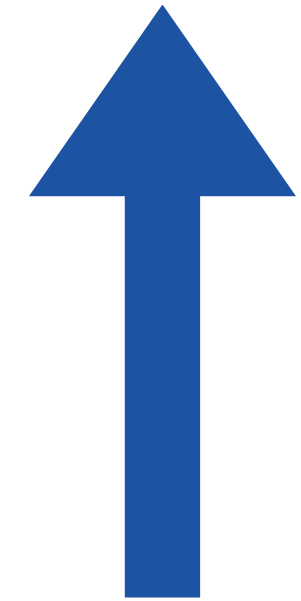


How to train a VAE?

$$\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - D_{KL}(q_{\phi}(z|x) \parallel p_{\theta}(z))$$

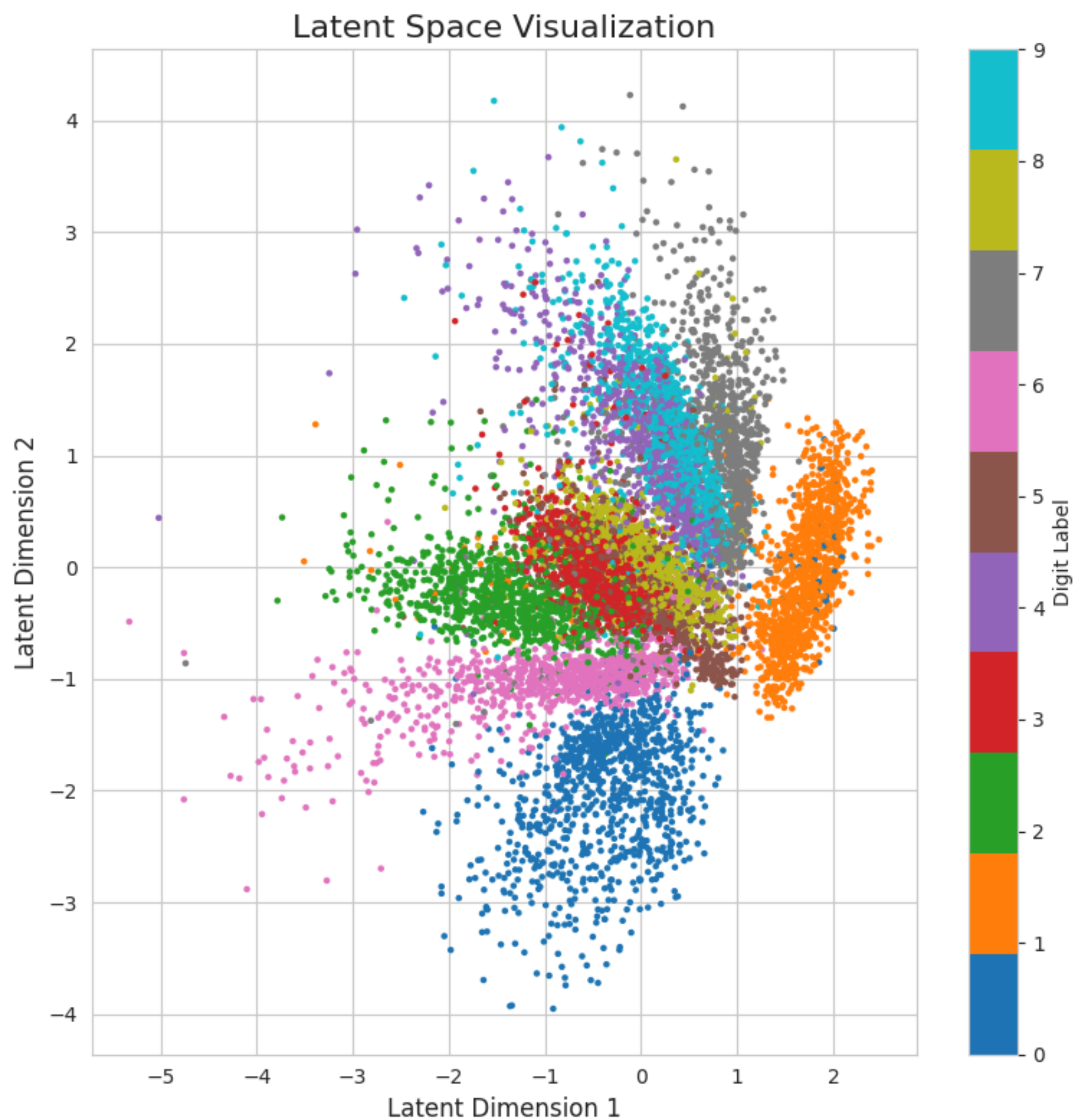
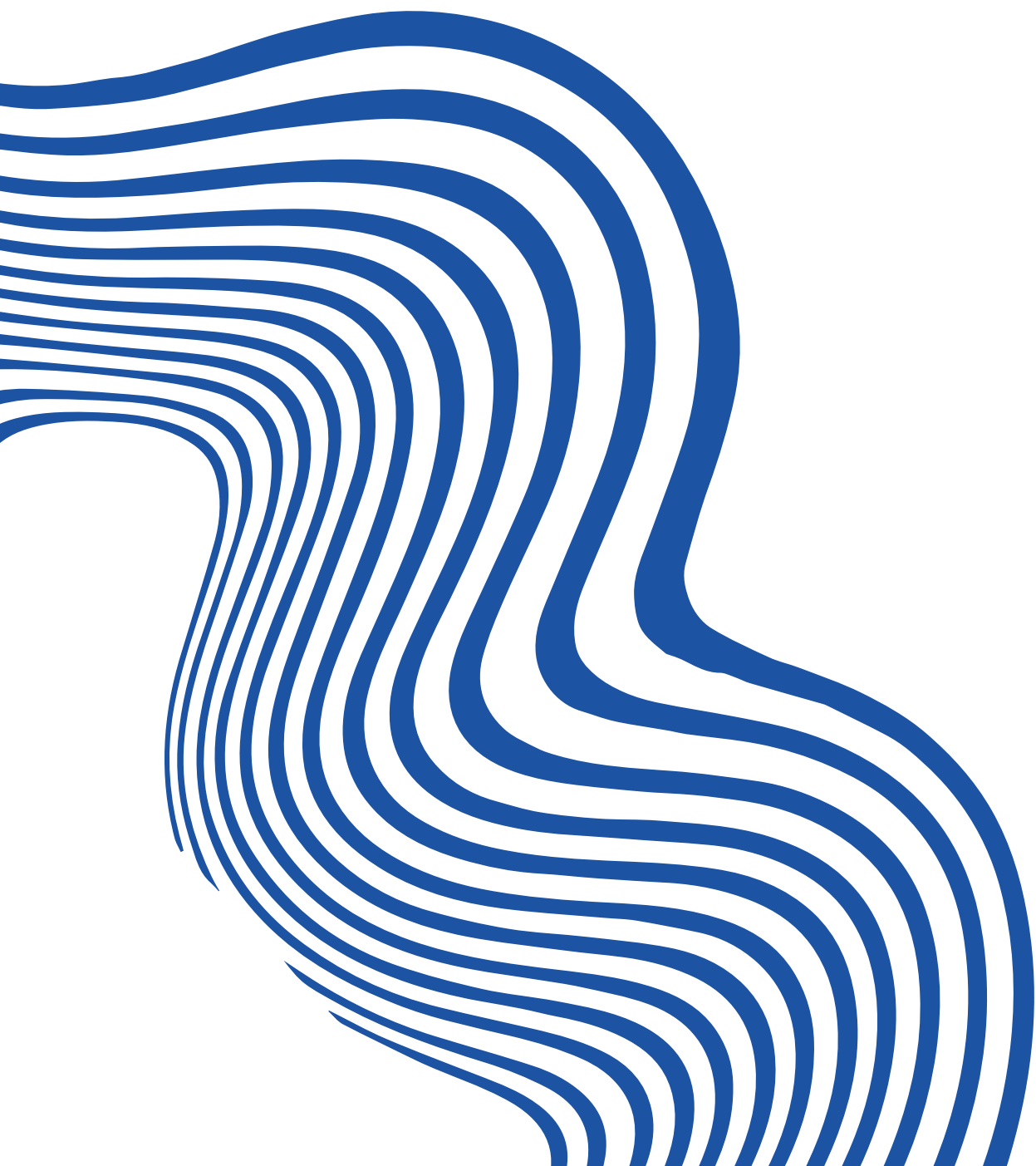


**Decoding error
(Cross-Entropy or MSE)**



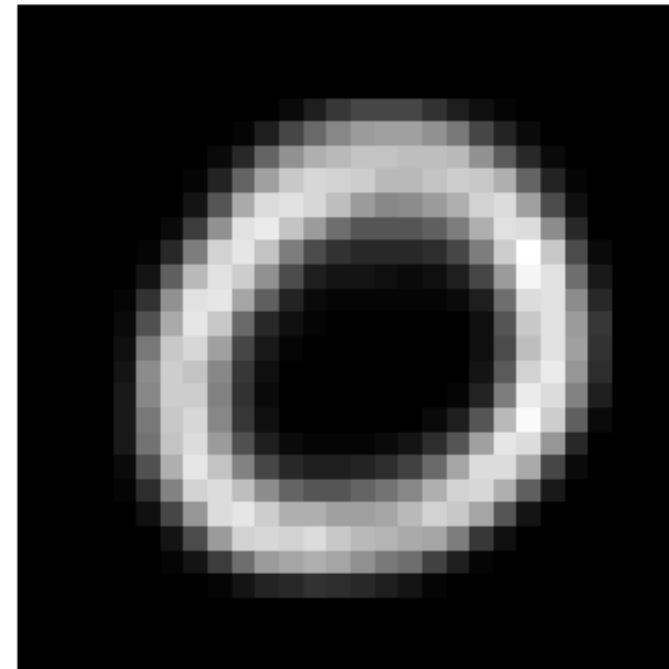
**Making the distribution
standard Gaussian**

Latent space

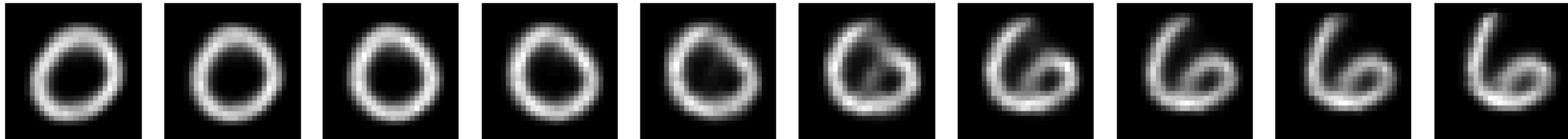
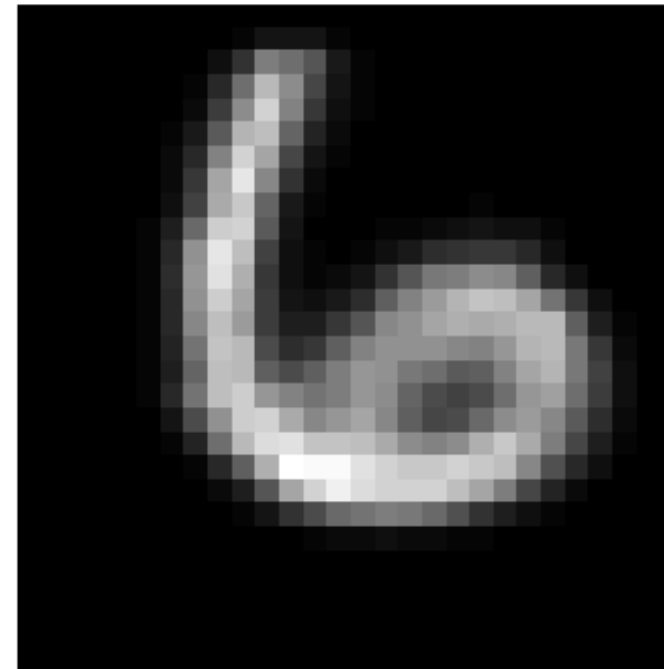


Convex interpolation

Test digit



Test digit



β -VAE

$$\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \beta D_{KL}(q_{\phi}(z|x) \parallel p_{\theta}(z))$$

t-Student VAE

$$\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - D_{KL}(q_{\phi}(z|x) \parallel p_{\theta}(z))$$

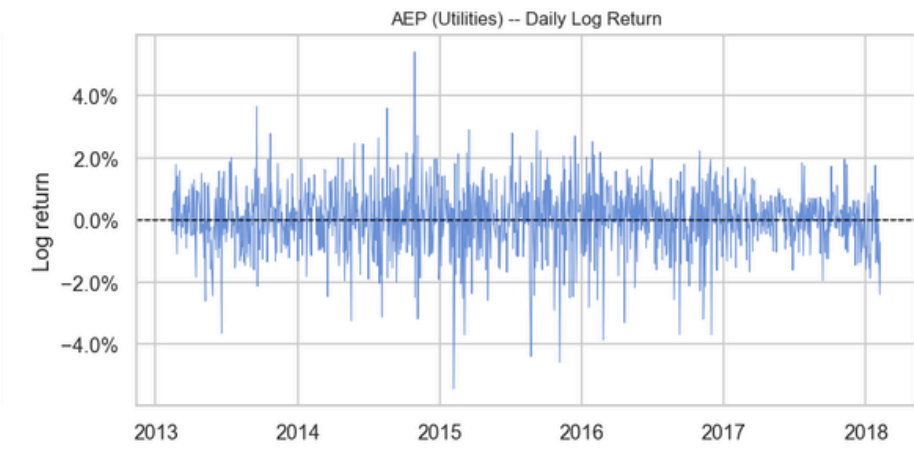
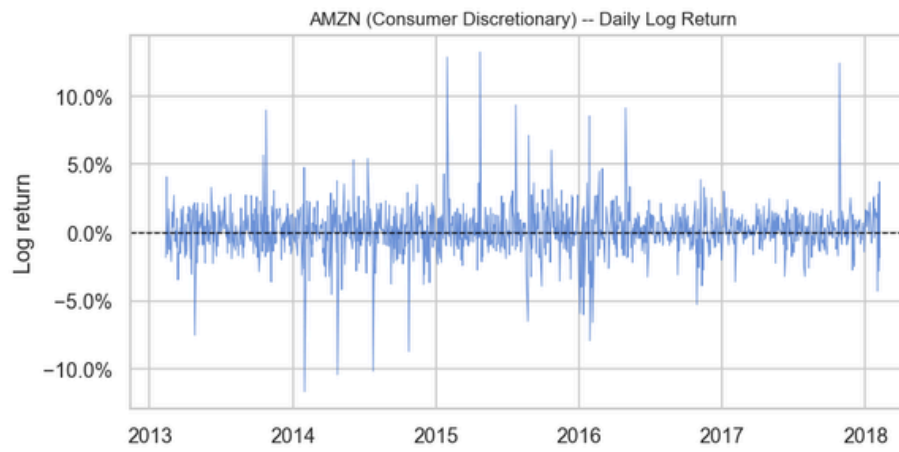
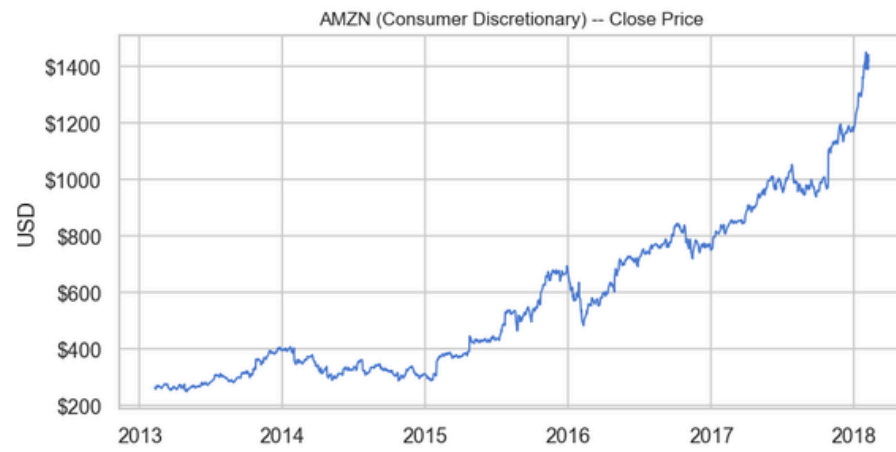
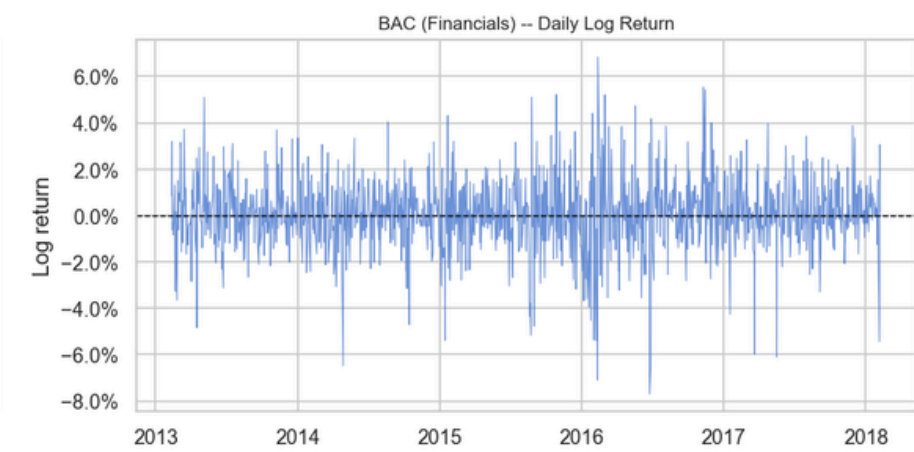
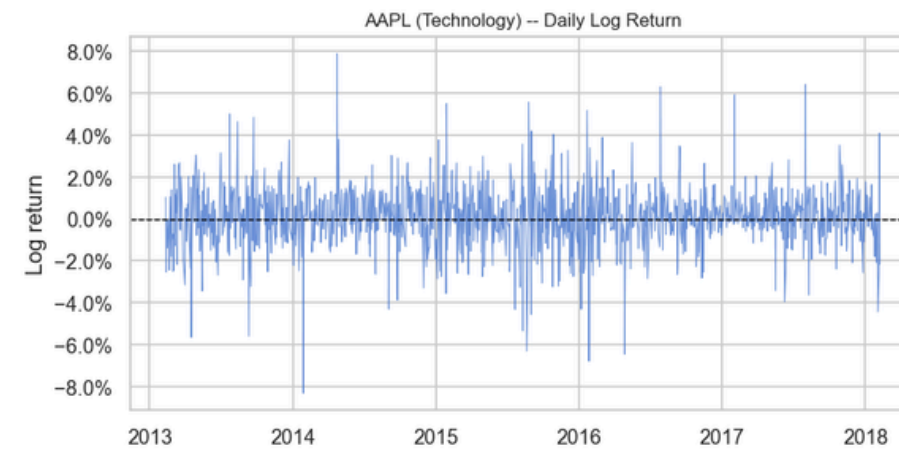
Data



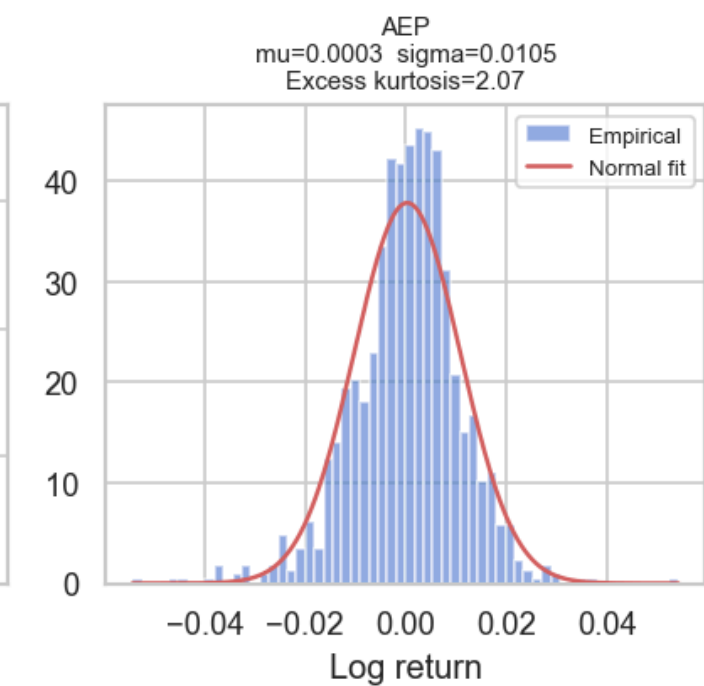
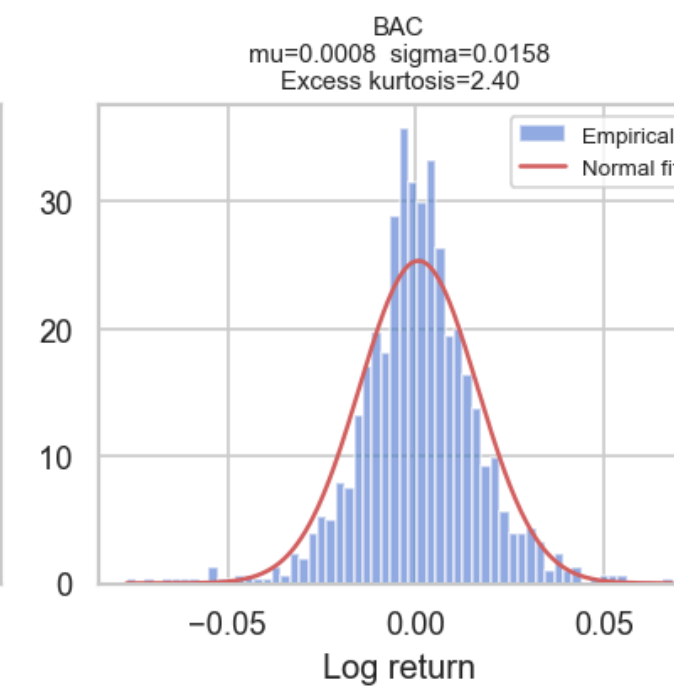
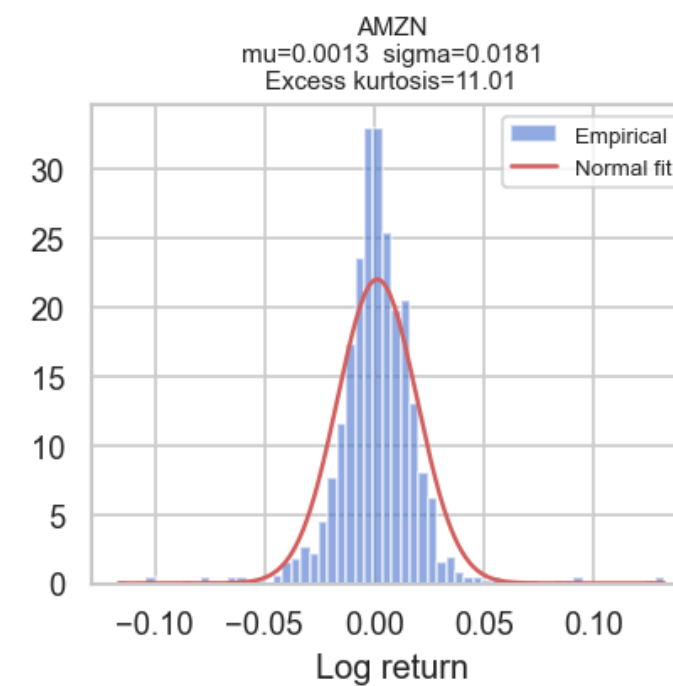
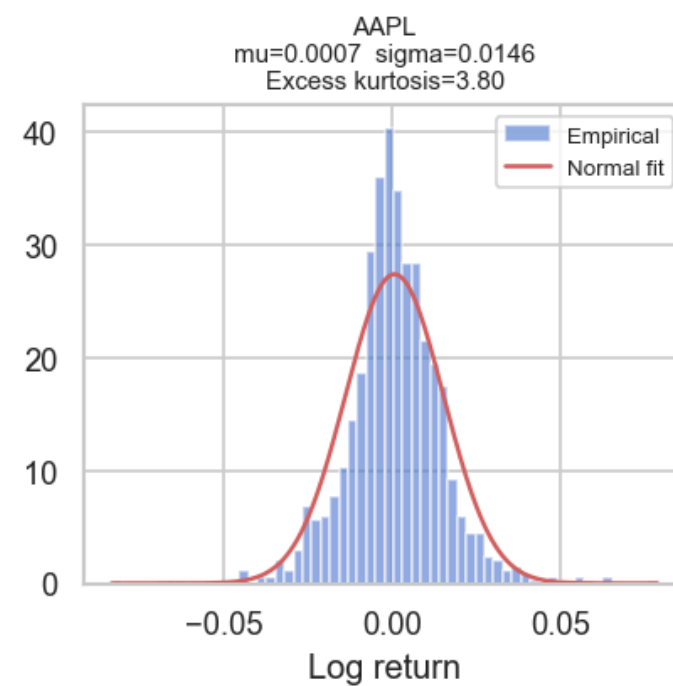
Nugent, C.: S&P 500 Stock Data. Kaggle dataset, historical stock data for all current S&P 500 companies (updated 2018). <https://www.kaggle.com/datasets/camnugent/sandp500> :contentReference[oaicite:0]index=0

- **505 stocks (470 without missing values)**
- **from 08.02.2013 to 07.02.2018 (1259 days)**

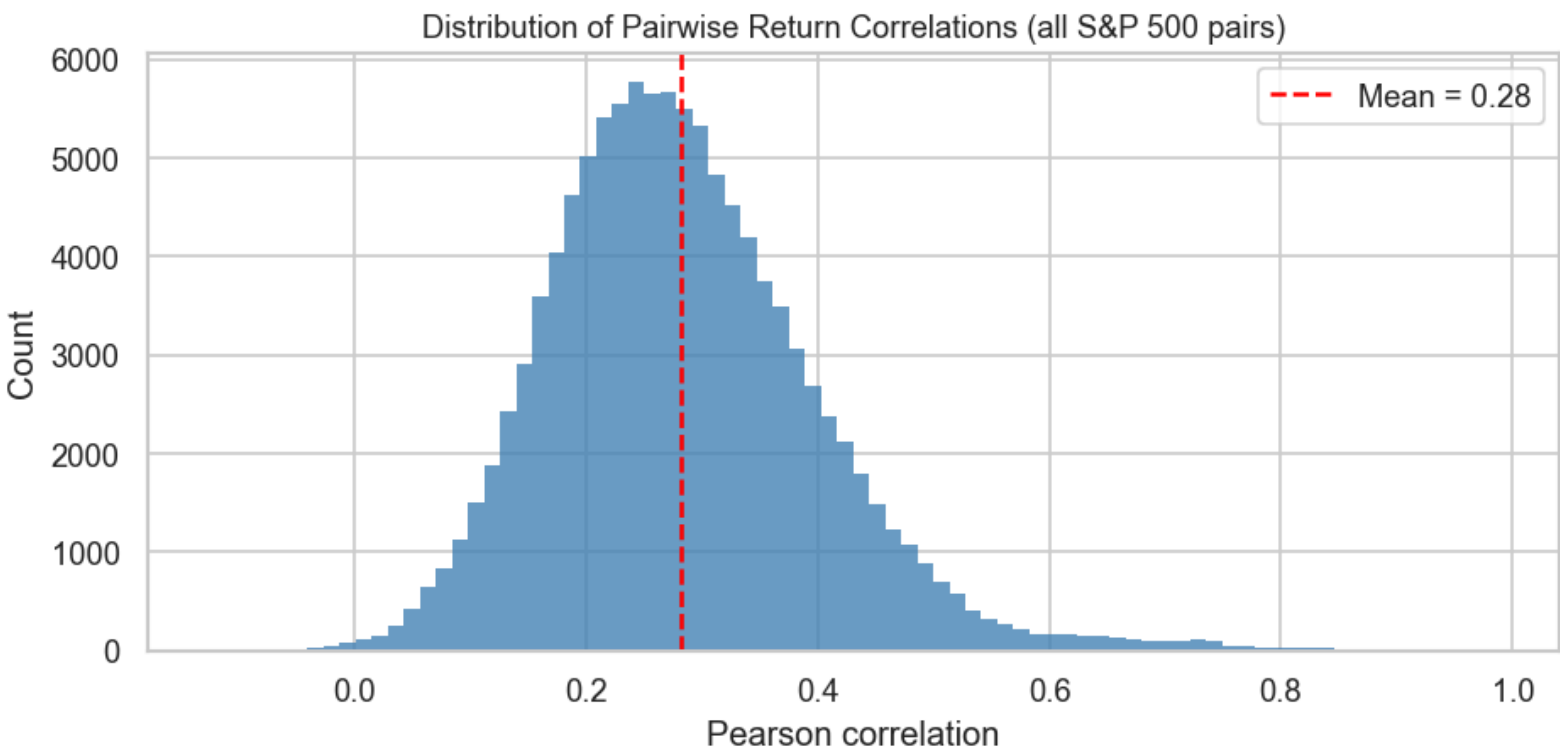
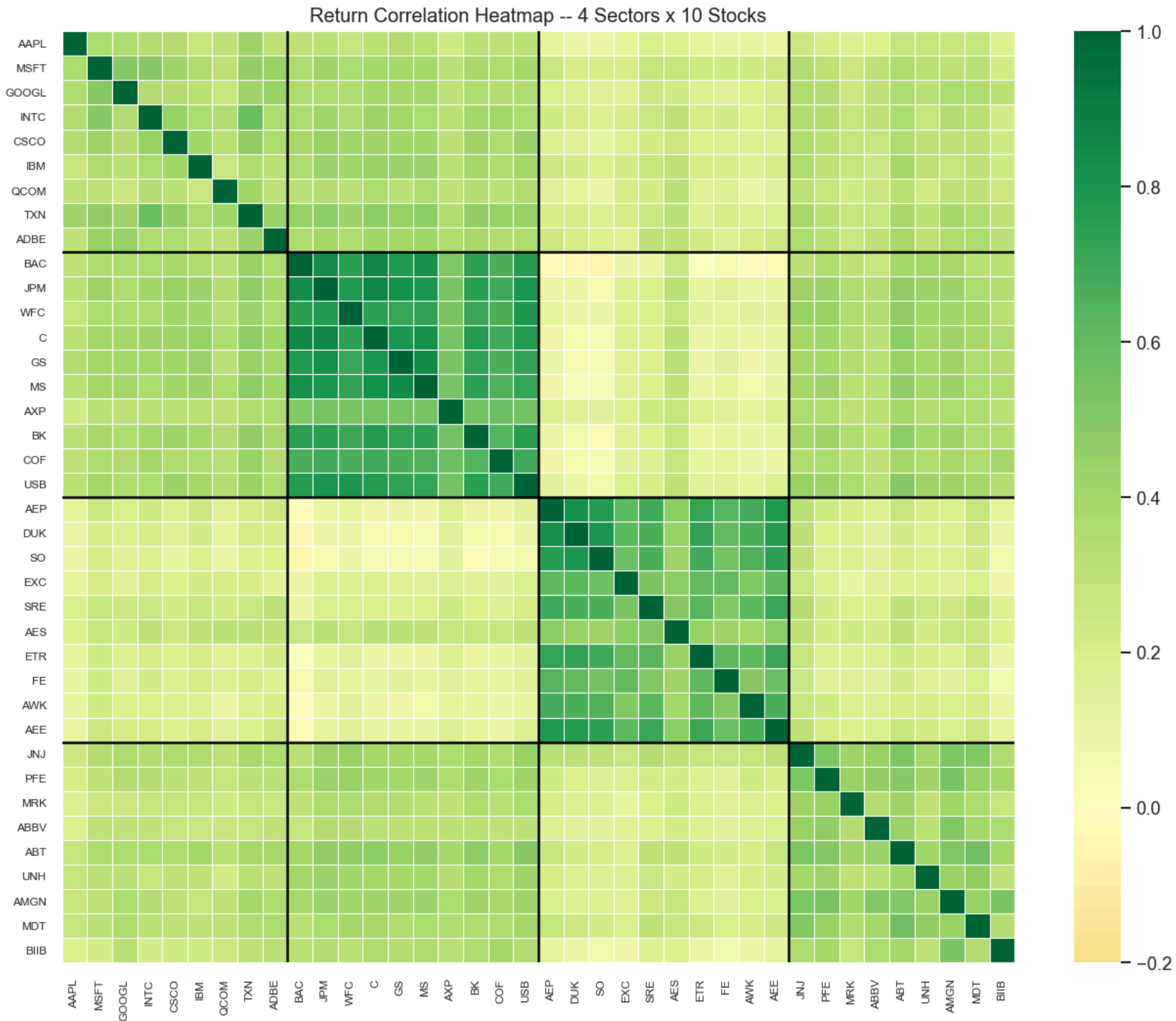
Data



Daily Log-Return Distributions

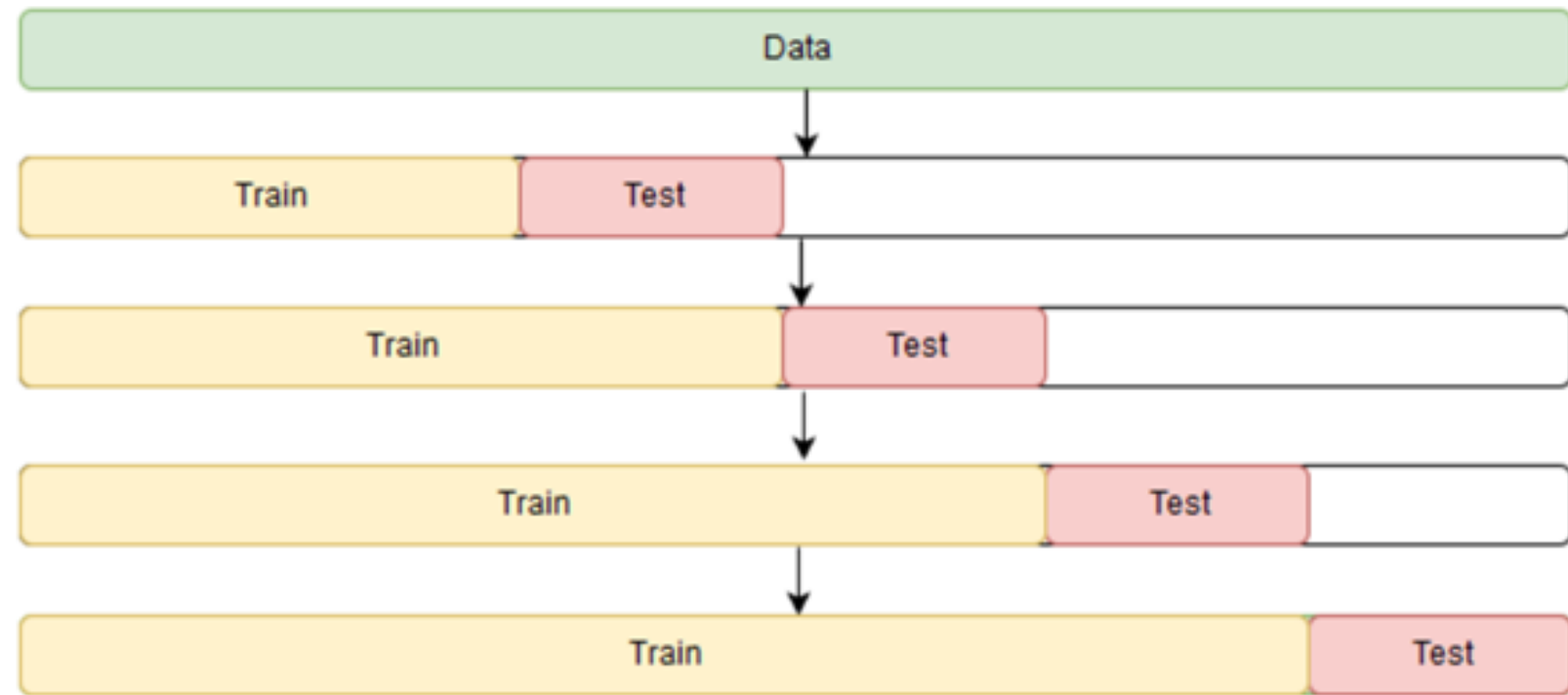


Data



Methodology

Training approach

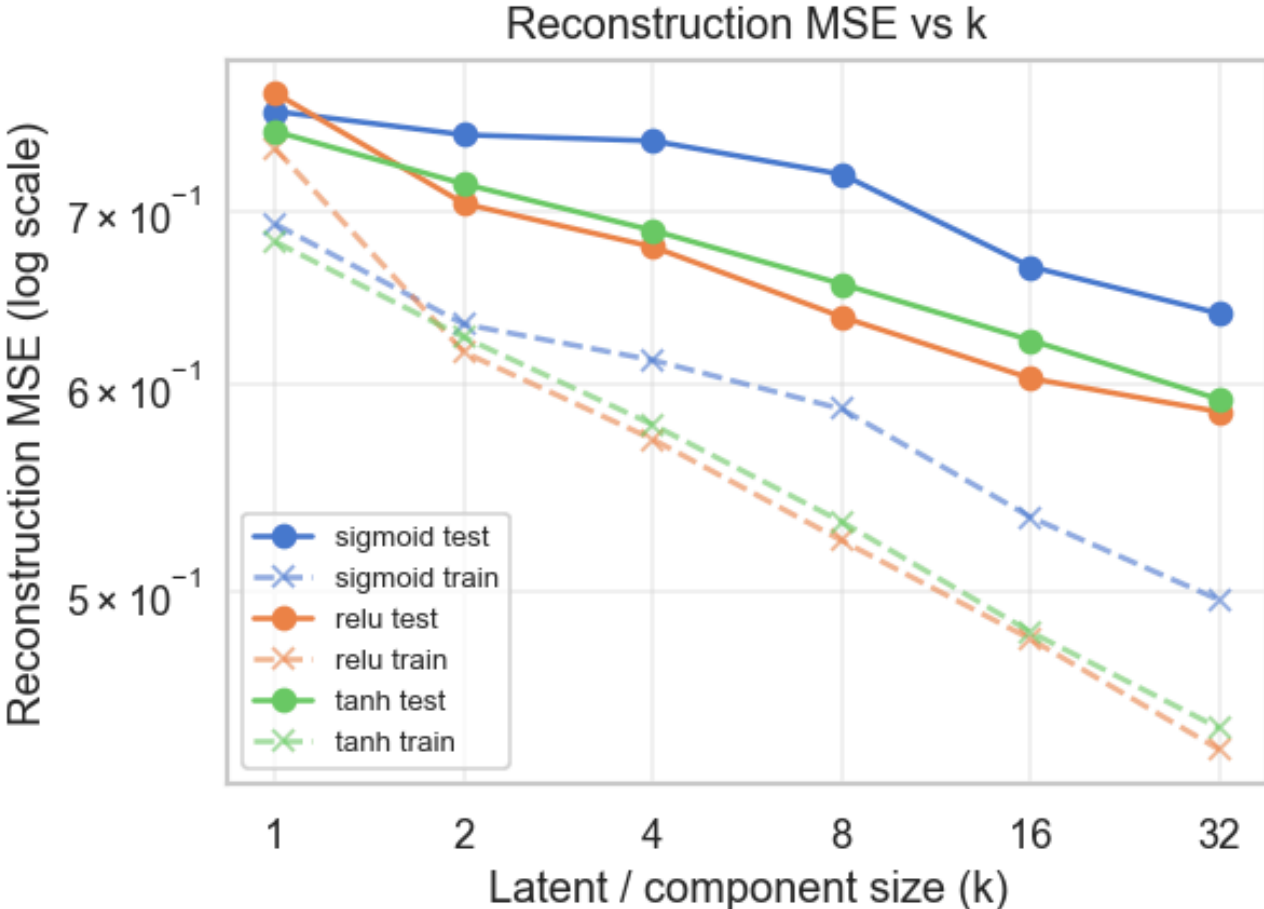
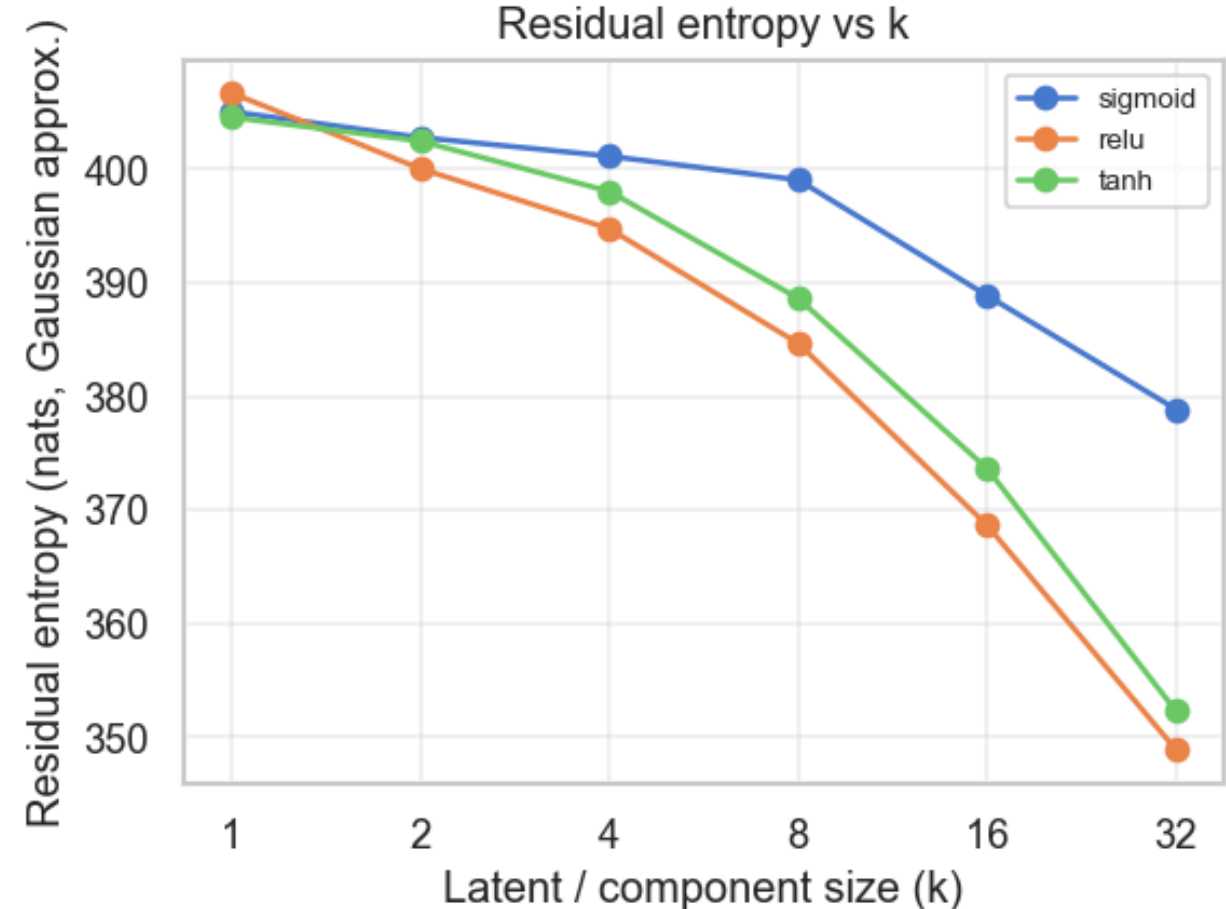
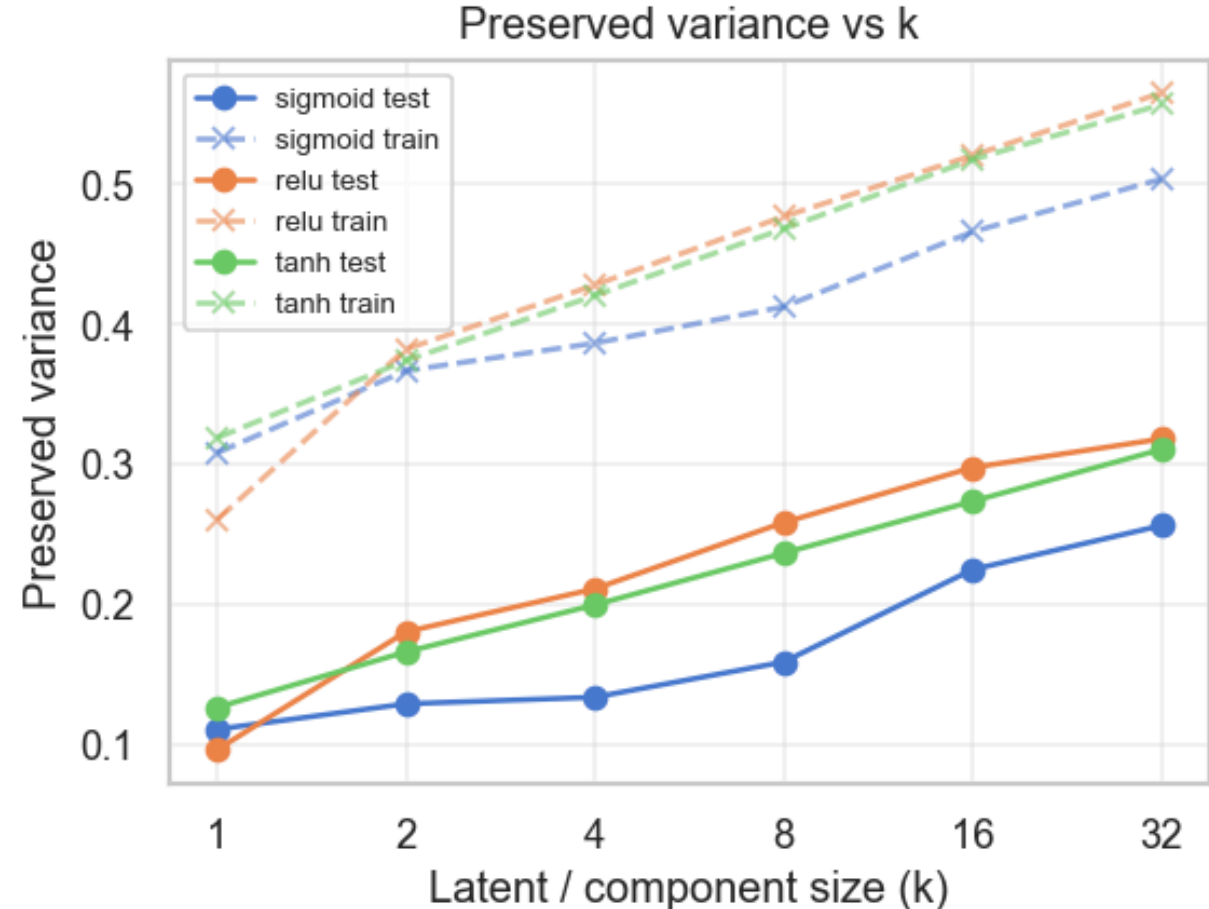


Metrics:

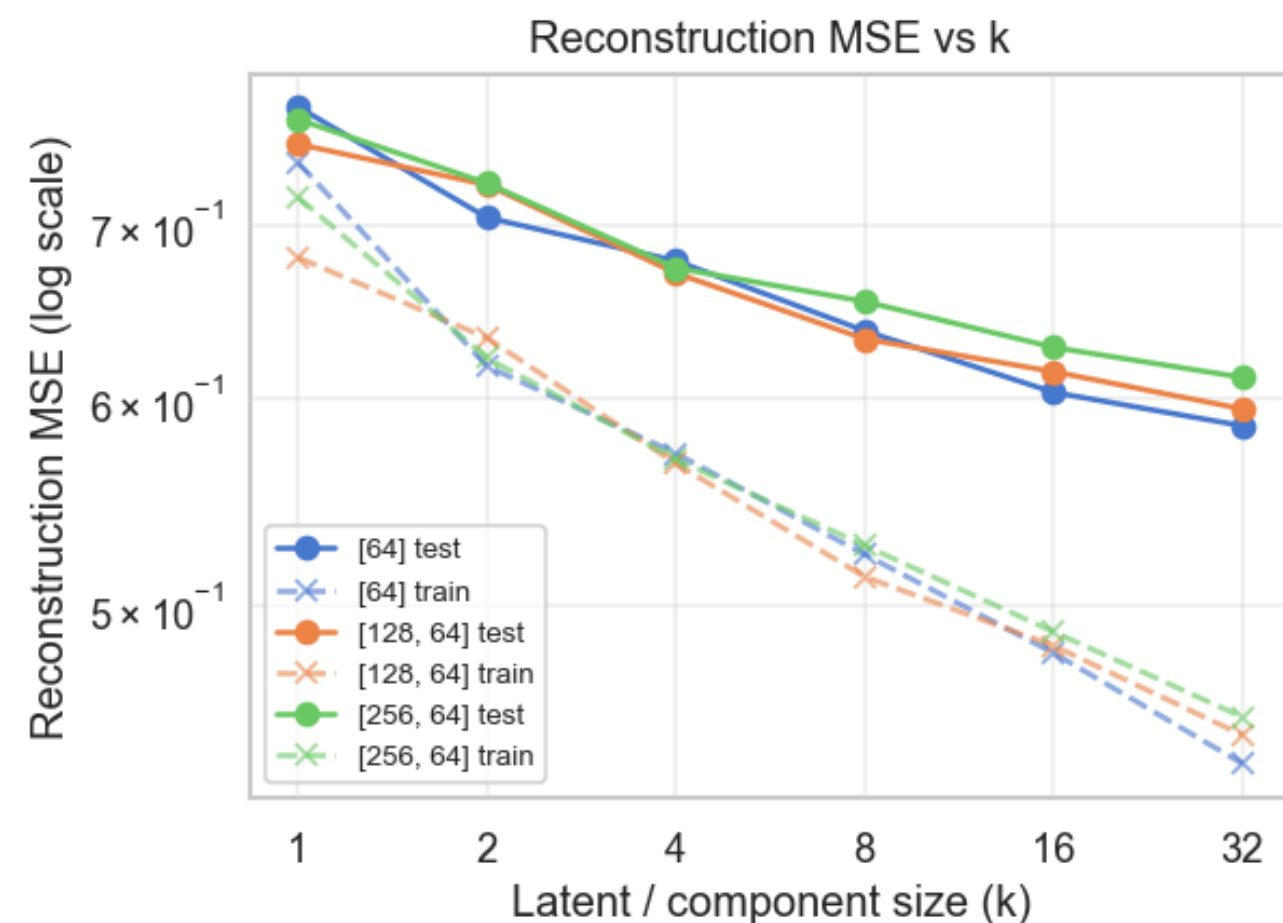
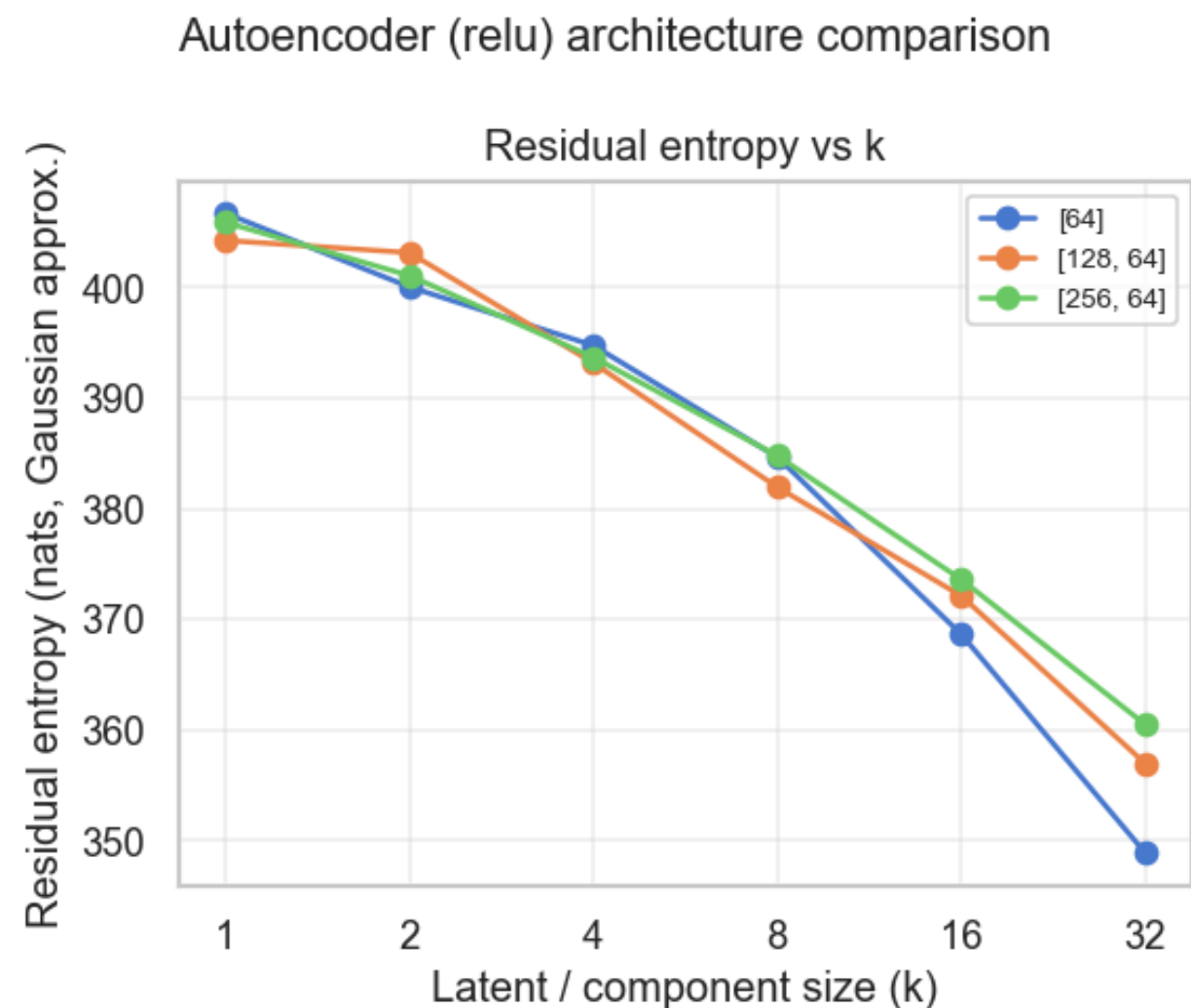
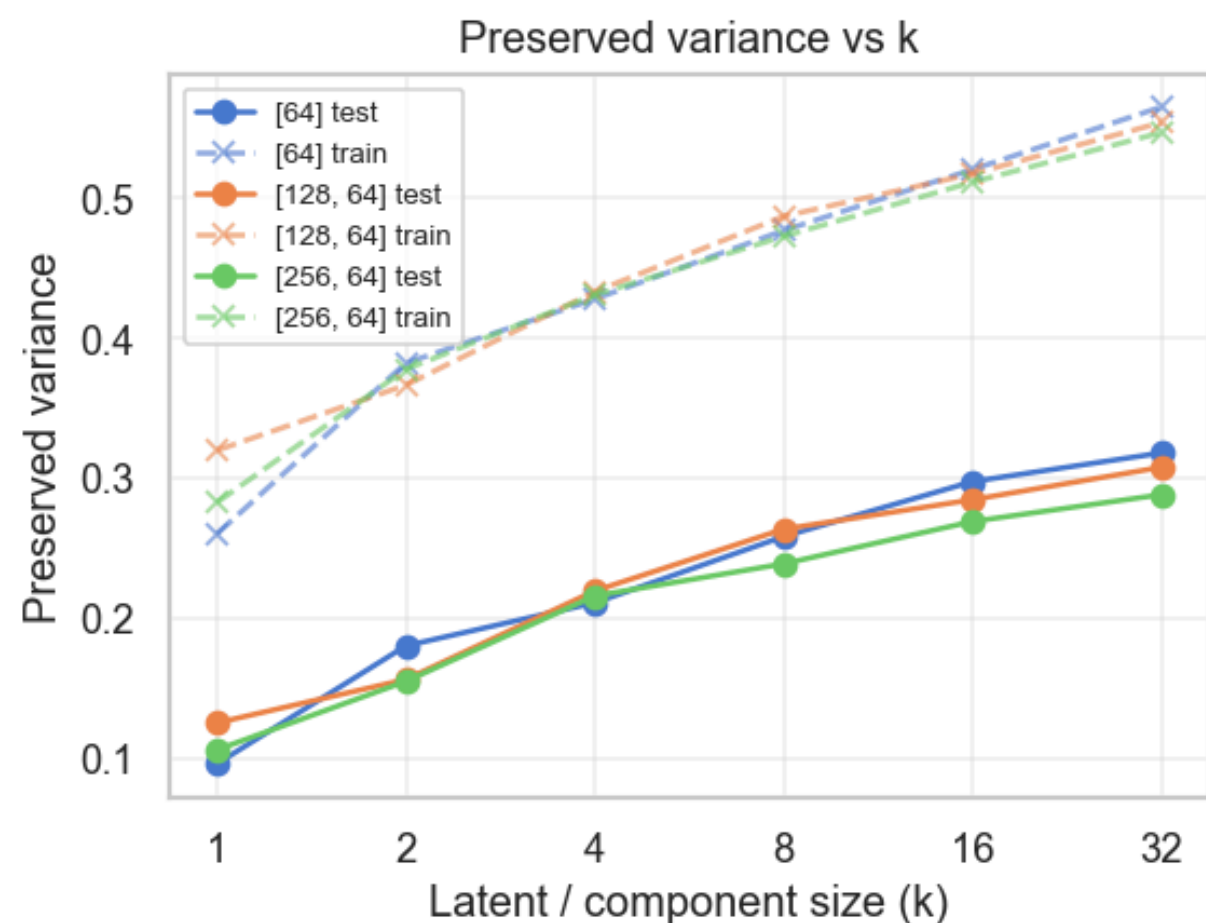
- Preserved variance
- Reconstruction error
- Residual entropy

Preliminary results

Autoencoder activation comparison -- sigmoid vs relu vs tanh

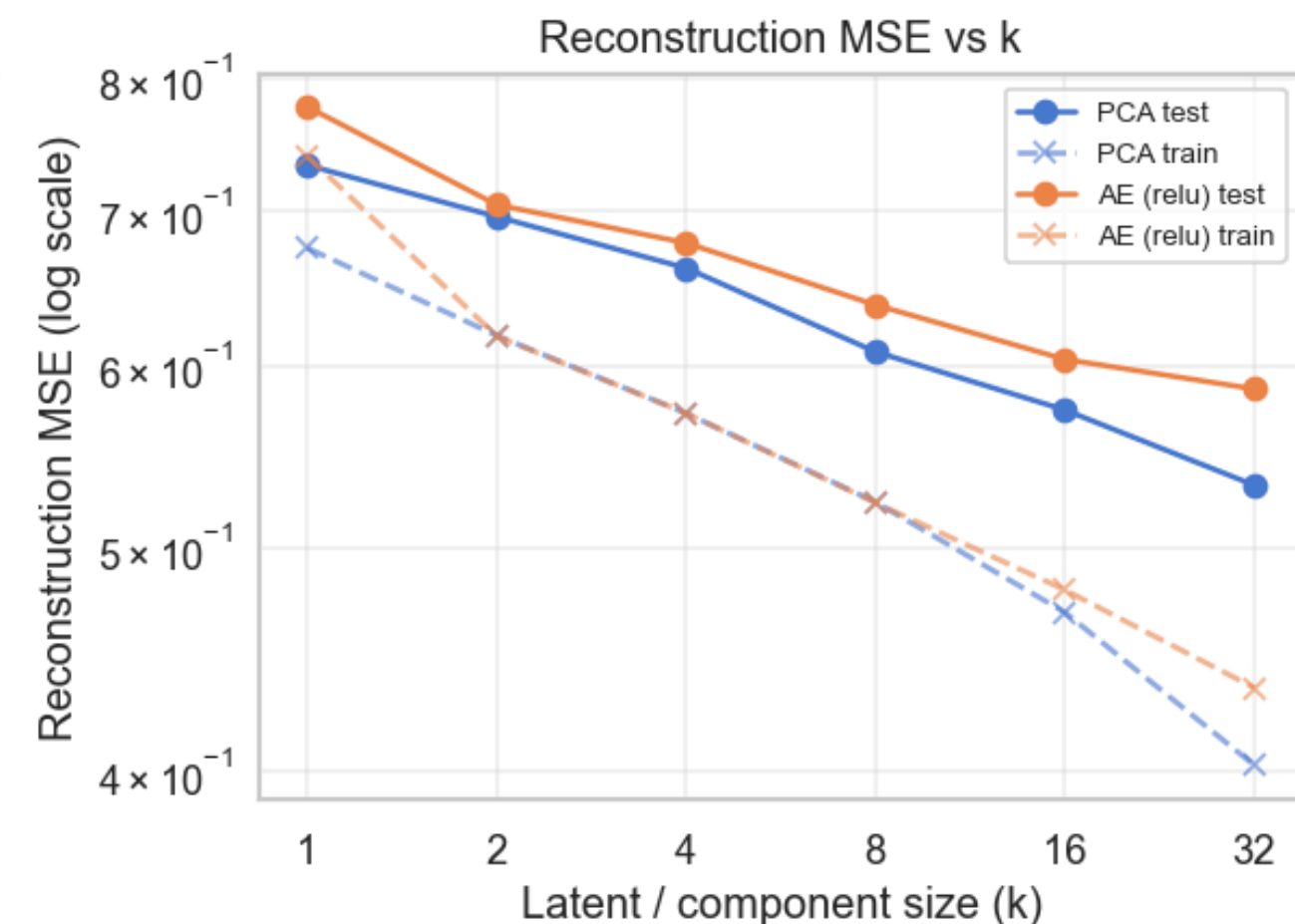
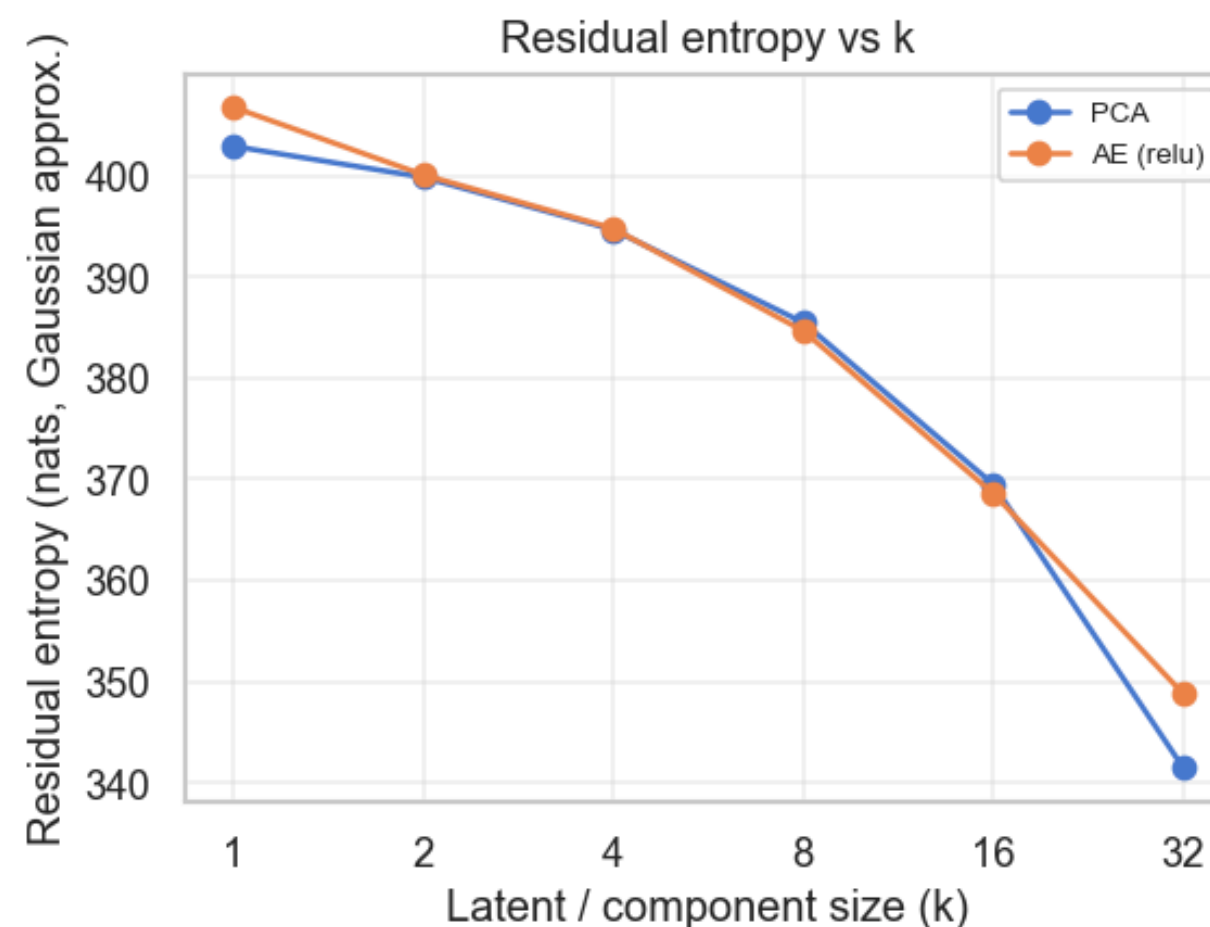
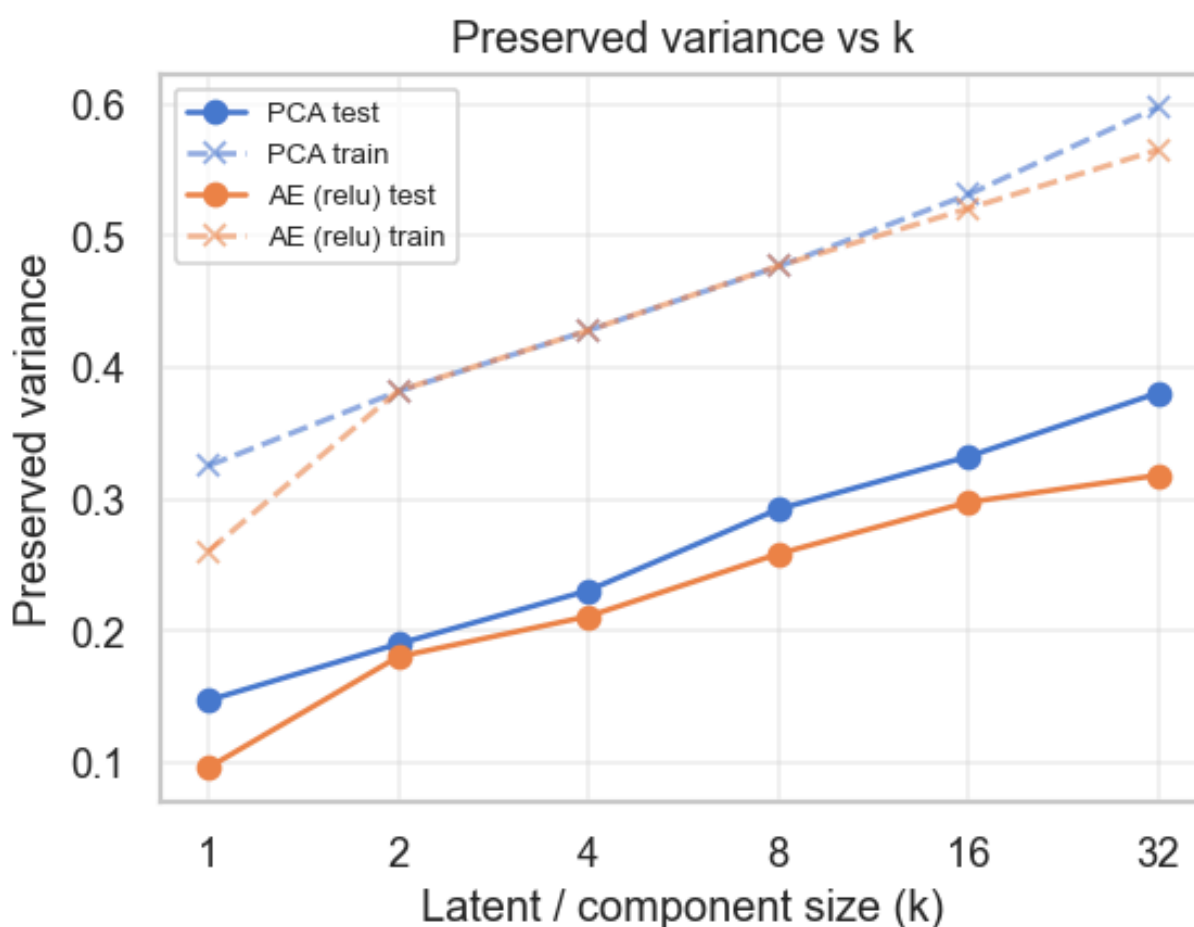


Preliminary results



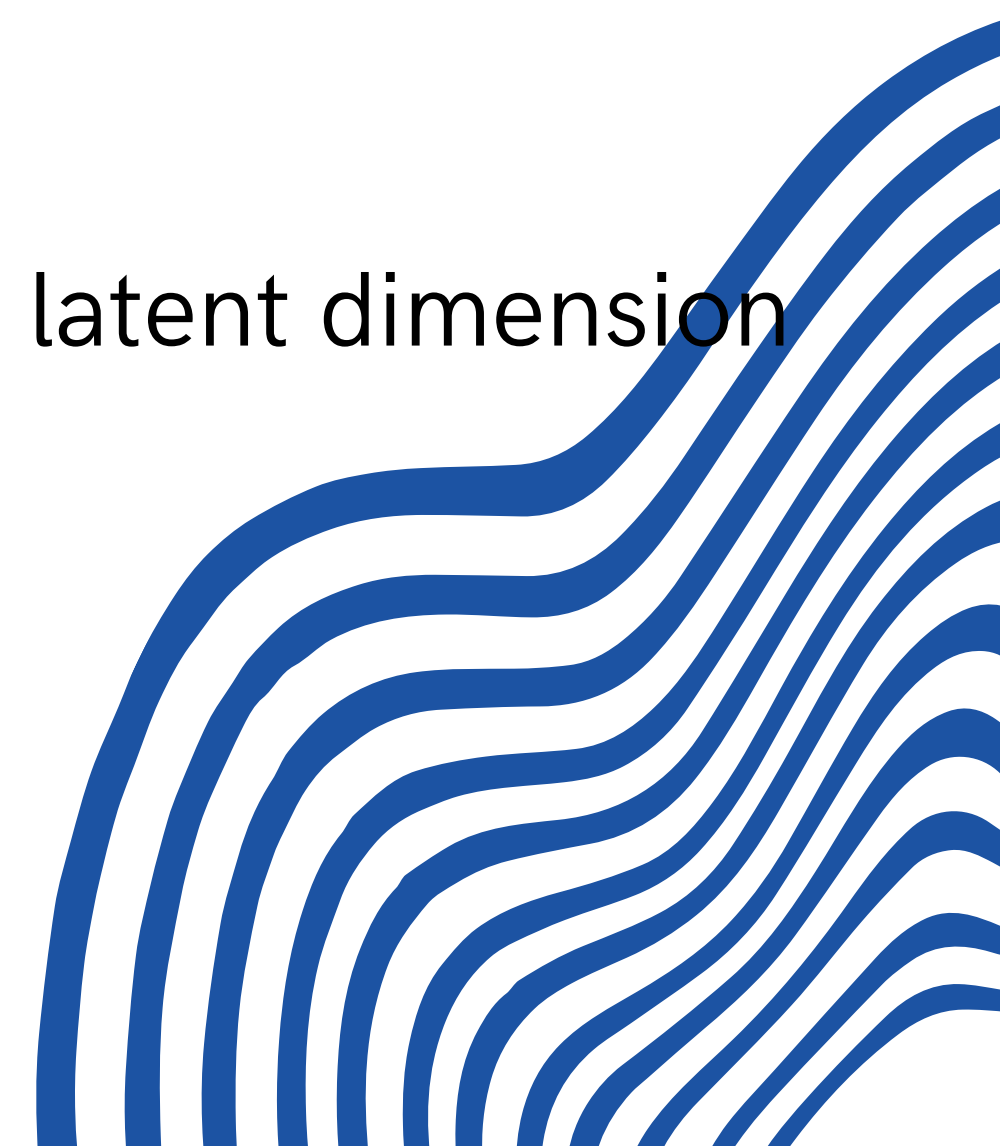
Preliminary results

PCA vs Autoencoder -- dimensionality-reduction metrics



Conclusion

- Regular autoencoder performance is slightly below the PCA
- The latent dimension structure might lead to some insights
- VAE should perform even worse in reconstruction
- VAE might have even more interesting structure of the latent dimension



Relevant literature

- 
- [1] Kingma, D.P., Welling, M.: Auto-Encoding Variational Bayes. In: 2nd International Conference on Learning Representations (ICLR), Conference Track Proceedings (2014). <https://arxiv.org/abs/1312.6114>
- [2] Song, T., Sun, J., Liu, X., Peng, W.: Scale-VAE: Preventing Posterior Collapse in Variational Autoencoder. In: Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING), pp. 14347–14357. ELRA and ICCL, Torino, Italia (2024). <https://aclanthology.org/2024.lrec-main.1250/>
- [3] Weng, L.: From Autoencoder to Beta-VAE. lilianweng.github.io (2018).
<https://lilianweng.github.io/posts/2018-08-12-vae/>
- [4] Gunduz, H. An efficient stock market prediction model using hybrid feature reduction method based on variational autoencoders and recursive feature elimination. Financial Innovation 7(1), 28 (2021).
<https://doi.org/10.1186/s40854-021-00243-3>
- [5] Cortés, D.G., Onieva, E., López, I.P., Trinchera, L., Wu, J.: Autoencoder-Enhanced Clustering: A Dimensionality Reduction Approach to Financial Time Series. IEEE Access 12, 16999–17009 (2024).
<https://doi.org/10.1109/ACCESS.2024.3359413>



Thank you!

Questions?

